Runp

runp contributors

Version 0.12.1

Overview

Like `docker-compose` but not just for containers.

Today, every non-trivial software project is compound of a number of parts (applications, libraries, services) running together in a lot of environments: virtual machines, Docker containers, physical boxes, cloud…

Runp allows you to specify what and how you want to run and helps you to start the system as a whole, running all needed processes in parallel (like Docker compose, Foreman et simili).

This speeds up the setup of a working environment compared to the usual approaches like custom scripts, documentation (written or spoken), or, more commonly, leaving it to the developer to figure out.

Runp uses a `Runpfile`, a sort of a sofisticated `Procfile` that describes the project as a system composed of `units` each of which can run a different type of process such as:

- processes on the physical box
- container processes
- SSH tunnel processes

Moreover the Runpfile handles:

- working directory per process
- environment variables per process
- user defined vars per system, defined in the Runpfile or passed at runtime as command line arguments
- processes dependency: a process can wait for a given condition to start, e.g. a file appearing or another process being ready
- explicit startup ordering via `depends_on`
- not only *long running* processes but *one shot commands* too
- Windows OS

An example of a `Runpfile` :

```
name: Example
description: |
  Sample Runpfile to show runp functionalities
units:
  web:
    description: Web app
    # this process is running on host machine
    host:
      command: node app.js
      workdir: backend
```

```
    env:
      # inherit PATH from host system to find needed tools (e.g. node)
      PATH: $PATH
    await:
      # wait for the DB being available
      resource: tcp4://localhost:5432/
      timeout: 0h0m10s
mail:
  description: Test mail server
  # this process is running in a container
  container:
    image: docker.io/mailhog/mailhog
    ports:
      - "8025:8025"
      - "1025:1025"
db:
  description: Corporate DB
  # This process is reachable through SSH port forwarding
  ssh_tunnel:
    user: user
    auth:
      identity_file: ~/.ssh/id_rsa
    local:
      port: 5432
    jump:
      host: dev.host
      port: 22
    target:
      host: corporate.db
      port: 5432
```

Commands

runp up

Start all units defined in the Runpfile. This is the **default command**: running `runp` with no subcommand is equivalent to `runp up`.

```
runp up
runp up --file ./infra/Runpfile
runp up --var DB_HOST=localhost --var DB_PORT=5432
runp up --key-env RUNP_SECRET_KEY
runp up --dry-run
```

Flags:

Flag	Description
`--file` / `-f`	Path to the Runpfile, or an `https://`/`http://` URL (see Remote Runpfiles). Overrides the `RUNP_FILE` environment variable. Default: `./Runpfile`.
`--checksum`	Expected SHA-256 checksum of a remote Runpfile, in the form `sha256:<64 hex characters>`. Required for `http://` URLs.
`--var` / `-V`	Set a runtime variable declared in the `vars:` section, e.g. `--var DB_HOST=localhost`. Can be repeated.
`--key-env`	Name of the environment variable that holds the decryption key for `encrypted_secret` fields (recommended).
`--key` / `-k`	Decryption key as a plain-text argument. **Visible in `ps aux` and shell history — prefer `--key-env`.**
`--dry-run` / `-n`	Print the units that would be started (name, kind, resolved command) without actually starting any process. Preconditions are evaluated without side effects: no container network is created and no SSH connection is opened.

`runp up` keeps running until all started processes have exited. On Ctrl-C (or `SIGTERM`) it stops all units, dependents before their dependencies. A second Ctrl-C (or `SIGTERM`) while units are being stopped forces runp to exit immediately; processes that have not stopped yet may keep running.

runp status (ps)

Show the runtime state of all units.

```
runp status
runp ps
runp status --file ./infra/Runpfile
```

Prints a table with columns NAME, KIND, STATE, and DETAIL. The `ps` alias is provided for convenience.

The state is probed from outside `runp up`, so `status` can run in another shell:

- host units: from the PID file written by `runp up` (a PID reused by another process is reported as stopped);
- container units: from the container engine;
- SSH tunnel units: by checking whether the local port accepts connections.

`--file` and `--checksum` work as for `runp up`.

runp list (ls)

List all units defined in the Runpfile.

```
runp list
runp ls
runp list --output json
runp list --output json | jq '.[].name'
runp list --file ./infra/Runpfile
runp list --var foo=bar
```

Flags:

Flag	Description
`--output` / `-o`	Output format. `table` (default) prints an aligned table with NAME, KIND, COMMAND, DESCRIPTION, PRECONDITIONS columns. `json` emits a JSON array of objects with the keys `name`, `kind`, `command`, `description` and `preconditions`, suitable for piping to `jq`.
`--var` / `-V`	Runtime variables in `key=value` format, as for `runp up`: the COMMAND column shows the commands with the vars resolved, so pass the same values used with `runp up`.

runp validate

Validate the Runpfile without starting any process.

```
runp validate
runp validate --file ./infra/Runpfile
```

Exit codes:

- `0` — file is valid
- `1` — validation errors found (missing process type, invalid unit name, undeclared variable,

unknown `depends_on` reference, cycle)

- `2` — file cannot be loaded (not found or invalid YAML)

Warnings do not change the exit code. They report OS or runp-version preconditions that would skip a unit on the current machine, and `await` blocks with a `resource` but no `timeout` (which are ignored).

runp reload

Stop and restart a single unit by name while the rest of the stack keeps running. Run this in a separate shell while `runp up` is active.

```
runp reload web
runp reload --file ./infra/Runpfile api
runp reload --var DB_HOST=localhost --key-env RUNP_SECRET_KEY api
```

NOTE Flags must be placed **before** the unit name: flags after it are ignored.

`reload` accepts the same `--file`, `--checksum`, `--var`, `--key` and `--key-env` flags as `runp up`; pass the same values so that the unit is restarted with the same configuration.

The running instance is stopped (host units through their PID file, container units through the container engine), then the unit is started again **in the shell running `reload`**, which keeps running until the unit exits. SSH tunnel units are not supported by `reload`.

runp encrypt

Encrypt a secret value for use as `encrypted_secret` in a Runpfile.

```
runp encrypt --key-env RUNP_SECRET_KEY mysecretvalue
echo mysecretvalue | runp encrypt --key-env RUNP_SECRET_KEY
runp encrypt mysecretvalue        # generates a random key and prints it with the
encrypted value
```

The secret can be supplied as a positional argument or piped via stdin. When neither is provided and stdin is a terminal, the command exits with an error.

WARNING

Avoid `--key` with a literal value — the key will be visible in `ps aux` output and recorded in your shell history. Use `--key-env` instead:

```
export RUNP_SECRET_KEY=my-encryption-key
runp encrypt --key-env RUNP_SECRET_KEY mysecretvalue
```

runp config

Manage user settings stored in `~/.runp/settings.yaml`.

Subcommand	Description
`runp config show`	Print the full contents of `~/.runp/settings.yaml` and its path.
`runp config get <key>`	Print the current value of a setting key (returns the default when not explicitly set).
`runp config set <key> <value>`	Write a setting key. Creates `~/.runp/settings.yaml` if it does not exist.

Example:

```
runp config show
runp config get container_runner
runp config set container_runner podman
```

Supported keys:

Key	Default	Description
`container_runner`	`docker`	Container runtime executable used for container units.

runp completion

Print the shell completion script for `bash`, `zsh` or `fish`. Completion covers subcommands, flags and, for `runp reload`, the unit names of the Runpfile.

```
source <(runp completion bash)                                  # in ~/.bashrc
source <(runp completion zsh)                                   # in ~/.zshrc, after
compinit
runp completion fish > ~/.config/fish/completions/runp.fish
```

Global flags

Flag	Description
`--log-level` / `-l`	Log verbosity. Accepted values: `trace`, `debug`, `info` (default), `warn`, `error`.
`--file` / `-f`	Path to the Runpfile for the default `up` command, e.g. `runp -f ./infra/Runpfile`.
`--no-color` / `-C`	Disable colored output. Also honoured via the `NO_COLOR` environment variable.
`--dry-run` / `-n`	Preview units without starting them (equivalent to `runp up --dry-run`).
`--debug` / `-d`	**Deprecated.** Equivalent to `--log-level debug`.
`--quiet` / `-q`	**Deprecated.** Equivalent to `--log-level warn`.

Exit codes

Code	Meaning
`0`	Success
`1`	`runp validate` found errors, or an unexpected internal error
`2`	The Runpfile cannot be loaded (not found, invalid YAML, invalid Runpfile, fetch failure)
`3`	Invalid arguments (e.g. mutually exclusive flags, missing unit name)
`4`	Variable error (e.g. `--var` for an undeclared variable, circular variable references)
`5`	Execution error (root preconditions not met, undeclared variable referenced by a unit, a unit failed to start or a dependency failed)
`128+N`	Stopped by signal `N`: `130` after Ctrl-C (`SIGINT`), `143` after `SIGTERM`

Usage

Run runp

Some commands:

```
runp --help                                # generic help
runp help up                               # describes the command "up"
runp up                                    # run the Runpfile in the current directory
(default command)
runp --log-level debug up -f /path/to/runpfile.yaml
runp encrypt --key-env RUNP_KEY secret     # encrypt "secret" using the key from
$RUNP_KEY
runp ls -f /path/to/runpfile.yaml          # list units in Runpfile
runp validate -f /path/to/runpfile.yaml    # validate Runpfile without starting
runp status                                # show runtime status of all units
```

Selecting the Runpfile

The Runpfile path is resolved in the following order of precedence:

1. `--file` / `-f` flag
2. `RUNP_FILE` environment variable
3. `./Runpfile` (default)

Example using the environment variable:

```
export RUNP_FILE=/path/to/my/Runpfile
runp up
runp validate
```

Remote Runpfiles

`--file` also accepts an `https://` or `http://` URL: the Runpfile is downloaded (at most 1 MiB, 30 seconds timeout) and verified against `--checksum` when given.

```
runp up --file https://example.com/project/Runpfile
runp up --file http://intranet/Runpfile --checksum sha256:<64 hex characters>
```

A Runpfile runs commands on your machine: plain `http://` URLs are accepted only together with `--checksum`. For the same reason, without `--checksum` a redirect from an `https://` URL to a non-`https://` URL is refused. Paths in a remote Runpfile (`include:`, `workdir`) are relative to a temporary directory, so remote Runpfiles should be self-contained.

Settings

Runp looks for a settings file in `~/.runp/settings.yaml`.

If the file does not exist or is empty/invalid, Runp starts using these defaults:

```
container_runner: docker
```

Use `runp config` to read or update settings without editing the file directly:

```
runp config set container_runner podman
runp config get container_runner
runp config show
```

Preconditions

Preconditions can be defined at two levels: at the root of the `Runpfile` or within a specific `unit`.

- **Root-level preconditions**: These are checked before any unit is executed. If any of these preconditions are not met, `runp` will exit and no units will be started. This is useful for ensuring that the environment is correctly set up before attempting to run any part of the system.
- **Unit-level preconditions**: These are checked before the specific unit is executed. If the preconditions for a unit are not met, that unit is skipped, but other units in the `Runpfile` will still be executed.

Available preconditions:

- **os**: This precondition checks if the operating system matches one of the specified values.
- **runp**: This precondition checks if the `runp` version satisfies the specified constraint.
- **hosts**: This precondition checks if `/etc/hosts` contains the given mappings.
- **env_vars**: This precondition checks environment variables.

os

The `os` precondition checks if the operating system is one of the specified `inclusions`.

Example:

```
preconditions:
  os:
    inclusions:
      - linux
      - darwin
```

In this example, the execution will only proceed if the operating system is Linux or macOS.

runp

The `runp` precondition checks if the `runp` version satisfies the specified constraint. The available operators are: `LessThan`, `LessThanOrEqual`, `Equal`, `GreaterThanOrEqual`, `GreaterThan`.

Example:

```
preconditions:
  runp:
    operator: GreaterThanOrEqual
    version: 0.10.0
```

This will ensure that `runp` version is at least `0.10.0`.

hosts

The `hosts` precondition checks for the presence of specific entries in the `/etc/hosts` file. `contains` maps an IP address to the host names that must be mapped to it.

Example:

```
preconditions:
  hosts:
    contains:
      127.0.0.1:
        - my.special.host
        - other.local
```

This precondition is satisfied if `/etc/hosts` maps both `my.special.host` and `other.local` to `127.0.0.1`.

env_vars

The `env_vars` precondition checks for the presence or value of environment variables. The available conditions are:

- `is_set`: the variable is set and not empty;
- `is_unset`: the variable is not set or is empty;
- `is_equal`: the variable has the given `value`.

Example of a root-level precondition:

```
preconditions:
  env_vars:
    - name: DATABASE_URL
      condition: is_set
```

In this case, `runp` will exit if the `DATABASE_URL` environment variable is not set.

Example of a unit-level precondition:

```yaml
units:
  my-service:
    preconditions:
      env_vars:
        - name: MY_SERVICE_ENABLED
          condition: is_equal
          value: "true"
    host:
      command: "echo 'My service is enabled'"
```

In this example, the `my-service` unit will only be executed if the `MY_SERVICE_ENABLED` environment variable is set to `"true"`.

Runpfile composition with `include:`

A Runpfile can include other Runpfiles using the `include:` key. Each entry is a path **relative to the directory of the file that declares it**. All units from included files are merged into the root Runpfile before startup.

```yaml
name: My Project
include:
  - infra/Runpfile      # defines units: db, redis
  - services/Runpfile   # defines units: api, worker
units:
  proxy:
    host:
      command: nginx -g "daemon off;"
```

After loading, `runp up` starts all units from all included files together.

The `vars` declared in included files are merged too, so they can be used (and set with `--var`) like vars of the root Runpfile; when the same var is declared in several files, the including file wins. Paths in each included file (e.g. `workdir`) are relative to the directory of that file.

Includes can be nested to any depth.

WARNING

Duplicate unit names across any two files are an error — loading fails with:

```
duplicate unit identifier: <name>
```

Circular includes are detected and reported with the full import chain:

```
circular dependency detected: a.yml -> b.yml -> c.yml -> b.yml
```

Including the same file twice (e.g. `a.yml` includes `b.yml` and `c.yml`, which both include `d.yml`) is not a cycle, but it defines the units of `d.yml` twice and is

> reported as a duplicate unit.

Startup ordering with `depends_on`

By default all units start concurrently. Use `depends_on` to declare explicit ordering when one unit must be ready before another begins.

```
units:
  db:
    ready:
      cmd: "pg_isready -h localhost"
    container:
      image: postgres:15
      ports: ["5432:5432"]
  redis:
    ready:
      delay: 2s
    container:
      image: redis:7
  migrate:
    depends_on: [db]
    host:
      command: ./migrate up
  api:
    depends_on: [migrate, redis]
    ready:
      output: "listening"
    host:
      command: ./api-server
  worker:
    depends_on: [api]
    host:
      command: ./worker
```

Each unit starts as soon as all the units it depends on are **ready**, independently of unrelated units:

- a unit with a `ready` condition (see below) is ready when the condition is satisfied;
- a unit without a `ready` condition is ready when its process **exits successfully**. This suits one-shot tasks such as `migrate`; a long-running service without a `ready` condition blocks its dependents until it stops, and runp logs a notice when this applies.

If a dependency fails — it cannot be started, its `ready` condition fails or times out, or (without a `ready` condition) it exits with an error — its dependents are not started, the units already running are stopped and runp exits with an error.

`runp validate` reports unknown references and circular dependencies as errors.

Units that are skipped due to unsatisfied preconditions are excluded from the graph; their dependents are treated as if the dependency is already satisfied.

Stop timeout

For every unit, you can specify a timeout to wait for the process to stop gracefully. If the process does not stop within the timeout, it will be killed. The default value is 5 seconds.

- host units: the process group of the unit (the shell and its children) receives `SIGTERM`, then `SIGKILL` after the timeout (on Windows the process is terminated);
- container units: the timeout is passed to `docker stop -t` (rounded up to whole seconds);
- SSH tunnel units are closed immediately.

```
units:
  zookeeper:
    description: Zookeeper
    stop_timeout: 0h0m15s
    container:
      name: zookeeper
      image: strimzi/kafka:0.19.0-kafka-2.5.0
      command: "bin/zookeeper-server-start.sh config/zookeeper.properties"
      ports:
        - "2181:2181"
      env:
        LOG_DIR: /tmp/logs
```

App waiting for another resource

A unit can wait for a resource to be available before starting. The `await` block can be used to specify a `resource` and a `timeout`.

`timeout` is required: an `await` block without `timeout` is ignored (`runp validate` warns about it). If the `resource` is not specified, the unit will wait for the given `timeout`. If the resource is not available within the timeout, the unit fails to start: as for any unit that cannot be started, runp stops the units already running and exits with an error.

Supported resources: `file:///path`, `tcp://host:port/` (also `tcp4`, `tcp6`), `unix:///path/to/socket`, `http://` and `https://` URLs.

```
units:
  app:
    description: app
    host:
      command: echo hi
      env:
        PATH: $PATH
      await:
        timeout: 0h0m03s
```

A backend app running on host waiting for a DB running in a container to be available:

```
units:
  be:
    description: Backend app
    host:
      command: mvn clean compile quarkus:dev
      workdir: backend
      env:
        # inherit PATH from host system to find mvn and java
        PATH: $PATH
      await:
        resource: tcp4://localhost:5432/
        timeout: 0h0m10s
  db:
    description: Database
    container:
      image: docker.io/postgres:alpine
      ports:
        - "5432:5432"
      env:
        POSTGRES_PASSWORD: pass
        POSTGRES_USER: user
        POSTGRES_DB: dbname
```

Wait for a file to be available:

```
units:
  await:
    description: command
    host:
      command: env
      await:
        resource: file:///tmp/some_file.txt
        timeout: 0h0m9s
      env:
        MYHOME: ${HOME}
```

Environment variables of runp (`$NAME` or `${NAME}`) are expanded in `env` values. Use `` ` for a literal `$` ``, e.g. `` `PASSWORD: paword `` sets `pa$word`.

Wait for an HTTP resource to be available:

```
units:
  run:
    description: docker from shell
    host:
      command: docker run -t --rm hello-world
      workdir: /tmp
      await:
```

```
        resource: http://localhost:8000
        timeout: 0h0m25s
  hello-web:
    description: crccheck/hello-world
    container:
      image: crccheck/hello-world
      ports:
        - "8000:8000"
```

Unit readiness conditions

By default, runp considers a unit ready (and starts its dependents) when its process exits successfully. For long-running services this is undesirable — dependents should start as soon as the service is up, not when it eventually stops.

The `ready` block configures a readiness condition that tells runp when a unit is ready for its dependents to proceed. Three condition types are supported; at most one should be set per unit. If multiple fields are set, `output` takes priority, then `cmd`, then `delay`.

ready.delay — wait a fixed duration after the process starts:

```
units:
  db:
    description: Database (ready after 3s warm-up)
    ready:
      delay: 3s
      timeout: 30s   # how long to wait before giving up (default: 30s)
    host:
      command: postgres -D /var/lib/pgsql/data
  api:
    description: App server (starts after db is ready)
    depends_on: [db]
    host:
      command: node server.js
```

ready.output — match a substring in the process stdout/stderr:

```
units:
  api:
    description: App server (ready when "listening" appears in output)
    ready:
      output: "listening"
      timeout: 30s
    host:
      command: node server.js
  worker:
    description: Background worker (starts after api is ready)
    depends_on: [api]
    host:
```

```
      command: node worker.js
```

ready.cmd — poll a shell command every second; ready when it exits 0. The command runs with the default shell (`bash` on Unix, `cmd` on Windows) in the working directory and with the environment of runp:

```
units:
  api:
    description: App server (ready when health endpoint responds)
    ready:
      cmd: "curl -sf http://localhost:8080/health"
      timeout: 60s
    host:
      command: ./api-server
  consumer:
    description: Consumer (starts after api health check passes)
    depends_on: [api]
    host:
      command: ./consumer
```

NOTE

`ready` and `await` serve complementary roles:

- `await` waits for an **external** resource before **this** unit's process starts (e.g. "wait for the DB port to be open before launching the app").
- `ready` determines when **this** unit signals to its **dependents** that they may proceed (e.g. "dependents start when this service prints 'listening'").

Both can be used together on the same unit.

If the `ready` condition of a unit fails or times out, the units depending on it are not started (see **Startup ordering with `depends_on`**); a unit nobody depends on keeps running.

Containers

You can set the container engine using the settings file (key: `container_runner`).

Example:

```
container_runner: /path/to/podman
```

WARNING

Only Docker and Podman (as they use the same command line flags) are supported.

Containers can talk to each other through a Docker network (`runp-network`), created automatically when the first container unit starts.

The container name (the host name exposed to other containers) is set to `runp-${UNIT_NAME}` or to

the field `name`.

Containers started by runp carry the label `runp.project` identifying their Runpfile: `runp reload` and `runp status` never act on a container with the same name started for another Runpfile. If a container with the unit's name already exists (running or stopped), the unit is not started and runp tells how to remove it.

The container `command` is split into arguments with shell quoting rules (quotes and backslashes) but is not expanded by a shell on the host: use `sh -c '…'` to run a shell inside the container.

This Runpfile starts Wordpress and MySql:

```
name: Wordpress Runpfile
description: Runpfile to run Wordpress and MySql
units:
  db:
    container:
      name: db
      image: docker.io/mysql:5.7
      ports:
        - "3306:3306"
      env:
        MYSQL_ROOT_PASSWORD: somewordpress
        MYSQL_DATABASE: wordpress
        MYSQL_USER: wordpress
        MYSQL_PASSWORD: wordpress
  wordpress:
    container:
      image: docker.io/wordpress:latest
      ports:
        - "8000:80"
      env:
        WORDPRESS_DB_HOST: db:3306
        WORDPRESS_DB_USER: wordpress
        WORDPRESS_DB_PASSWORD: wordpress
        WORDPRESS_DB_NAME: wordpress
      await:
        resource: tcp4://localhost:3306/
        timeout: 0h0m20s
```

Use containers volumes

Run containers and volumes (example is from the book Docker in action - Manning):

```
name: Containers Runpfile
description: This is Runpfile showing Docker volumes
units:
  fowler:
    description: The Fowler collection
```

```
    container:
      image: docker.io/alpine:3.12
      skip_rm: true
      mounts:
        - "type=volume,dst=/library/PoEAA"
        - "type=bind,src=/tmp,dst=/library/DSL"
      command: |
        echo "Fowler collection created"
  knuth:
    description: The Knuth collection
    container:
      image: docker.io/alpine:3.12
      skip_rm: true
      mounts:
        - "type=volume,dst=/library/TAoCP.vol1"
        - "type=volume,dst=/library/TAoCP.vol2"
        - "type=volume,dst=/library/TAoCP.vol3"
      command: |
        echo "Knuth collection created"
  reader:
    description: The avid reader
    container:
      image: alpine:3.12
      volumes_from:
        - fowler
        - knuth
      command: |
        ls -l /library/
      await:
        timeout: 0h0m3s
```

On Windows

Windows is supported:

```
name: Test Runpfile
description: This is Runpfile
units:
  await:
    description: read environment variables
    host:
      command: set
      env:
        # in env block variables have always the unix notation
        MYHOME: ${HOME}
  echo:
    description: echo the value of %OS% env var
    host:
      # when used in command, env vars have the specific OS notation
      command: echo %OS%
```

```
  infiniteloop:
    description: infinite loop
    host:
      # this script is in examples/ directory
      executable: infinite.cmd
      workdir: examples
```

Some programs, especially on Windows, implicitly use OS specific environment variables.

If you run into strange problems try adding these to the env block:

```
# Windows env vars
SystemRoot: ${SystemRoot}
ALLUSERSPROFILE: ${ALLUSERSPROFILE}
APPDATA: ${APPDATA}
CommonProgramFiles: ${CommonProgramFiles}
CommonProgramW6432: ${CommonProgramW6432}
ComSpec: ${ComSpec}
DriverData: ${DriverData}
HOMEDRIVE: ${HOMEDRIVE}
HOMEPATH: ${HOMEPATH}
LOCALAPPDATA: ${LOCALAPPDATA}
OS: ${OS}
PATHEXT: ${PATHEXT}
ProgramData: ${ProgramData}
ProgramFiles: ${ProgramFiles}
ProgramW6432: ${ProgramW6432}
PSModulePath: ${PSModulePath}
PUBLIC: ${PUBLIC}
SESSIONNAME: ${SESSIONNAME}
SystemDrive: ${SystemDrive}
TEMP: ${TEMP}
TMP: ${TMP}
USERNAME: ${USERNAME}
USERPROFILE: ${USERPROFILE}
windir: ${windir}
```

Specify a shell for host processes

For host processes, you can specify a shell to execute the command. By default, `runp` runs `command` with `bash --noprofile --norc -eo pipefail -c` on Unix (so `bash` must be installed) and with `cmd /C` on Windows. The command is appended to `args` as the last argument.

Use `executable` and `args` instead of `command` to run a program directly, without a shell.

```
units:
  bash-process:
    description: a process that runs with bash
    host:
```

```
      command: "echo $BASH_VERSION"
      shell:
        path: /bin/bash
        args:
          - -c
```

Run a different command on different operative systems

Inclusions are compared to `runtime.GOOS`:

```
units:
  win:
    description: Windows unit
    preconditions:
      os:
        # this unit is ran when os is windows
        inclusions:
          - windows
    host:
      command: dir {{vars runp_root}}
  unix:
    description: Nix unit
    preconditions:
      os:
        # this unit will be ran when os is linux or darwin
        inclusions:
          - linux
          - darwin
    host:
      command: ls -al {{vars runp_root}}
```

SSH tunnel to reach a remote LDAP

A backend app running on host using LDAP on remote server available using SSH tunneling.

SSH tunnel can use three auth methods:

- `identity_file`: the path to the private key, ie `~/.ssh/id_rsa`
- `secret`: the SSH server password in plain text
- `encrypted_secret`: the SSH server password encrypted and in base 64 (you can create it using `runp encrypt`)

The host key of the jump server is verified against `~/.ssh/known_hosts` (or the file set in `known_hosts_file`): add the server key there before starting the tunnel, e.g. by connecting once with `ssh`.

`test_command` (optional) is run on the jump server before the tunnel starts; if it fails, the unit is skipped.

Each connection to the local port is forwarded independently through a single SSH connection to the jump server.

```
units:
  be:
    description: Backend app
    host:
      command: mybackendapp
      workdir: backend
  ldap:
    description: LDAP
    ssh_tunnel:
      user: runp
      auth:
        #identity_file: ~/tmp/runpssh/ssh/runp
        #secret: "plain text secret"
        encrypted_secret: "NsM1hcAy/L2TfACgfzbhYyb9j5a2ySYcARFDKkv7HTk="
      local:
        # localhost is the default
        port: 389
      jump:
        host: sshserver
        port: 22
      target:
        host: ldapserver
        port: 389
      # optional
      known_hosts_file: ~/.ssh/known_hosts
      test_command: "hostname"
```

WARNING

Setting `insecure_ignore_host_key: true` on an SSH tunnel unit disables host key verification and makes the connection vulnerable to man-in-the-middle attacks. Runp always prints a warning to stderr when this option is active, regardless of the `--log-level` setting. Use it only in controlled environments and never in production.

```
ssh_tunnel:
  insecure_ignore_host_key: true   # WARNING: MITM risk
  ...
```

Use secrets

SSH tunnel process allows user to use secrets to specify the password.

To create the encrypted secret:

```
export RUNP_SECRET_KEY=thekey
```

```
runp encrypt --key-env RUNP_SECRET_KEY SECRET
```

The above command will encrypt the string `SECRET` using the key stored in `$RUNP_SECRET_KEY`.

NOTE

You can also pipe the secret from stdin to avoid it appearing in your shell history:

```
echo SECRET | runp encrypt --key-env RUNP_SECRET_KEY
```

If no key is provided at all, `runp encrypt` generates a random key and prints it alongside the encrypted value.

To run a Runpfile containing an `encrypted_secret` you have to pass the key to the `up` command (the key must coincide with the one used to encrypt).

Use `--key-env` to pass the key through an environment variable (recommended):

```
export RUNP_SECRET_KEY=thekey
runp up --key-env RUNP_SECRET_KEY
```

WARNING

The `--key` / `-k` flag passes the key as a plain-text command-line argument. It will be visible in `ps aux` output and recorded in your shell history. Always prefer `--key-env`.

Use environment variables

Host processes do **not** inherit the environment of runp: they receive only the variables listed in their `env` block. Forward the variables a process needs explicitly, e.g. `PATH: $PATH` or `HOME: ${HOME}`. Values are expanded against the environment of runp; use `$$` for a literal `$`.

Container units receive their `env` block as `-e NAME=value` options; the container image defines the rest of their environment.

A one-shot command using custom environment variables:

```
env3:
  description: echo command
  host:
    command: echo ${MYHOME}
    workdir: ..
    env:
      MYHOME: ${HOME}
```

User defined variables

Use runtime vars:

```
vars:
  foo: FOO_DEFAULT_VALUE
units:
  vars-test-unit:
    description: echo a user defined var
    host:
      command: echo __{{vars foo}}__
```

Variable names may contain letters, digits and underscores (e.g. `DB_HOST`). Referencing a variable that is not declared is an error: runp does not start any unit.

The var `foo` will have value `FOO_DEFAULT_VALUE` or can be set from command line:

```
$ bin/runp --log-level debug up -f examples/Runpfile-vars.yml --var foo=bar
```

Implicit variables

Runp adds to the context some variables:

- `runp_workdir`: user current working directory as absolute path
- `runp_root`: directory parent of the Runpfile as absolute path
- `runp_file_separator`: OS file separator (`/` on unix, `\` on windows)

Usage:

```
units:
  vars:
    description: echo implicit vars from Runp
    host:
      command: "echo runp_workdir={{vars runp_workdir}} runp_root={{vars runp_root}}"
```

Disabling color output

To have plain, non-colored text output set the environment variable `NO_COLOR`:

```
NO_COLOR=1 ./bin/runp --log-level debug up -f examples/Runpfile-many-units.yml
```

or use the option `--no-color`:

```
./bin/runp --log-level debug --no-color up -f examples/Runpfile-many-units.yml
```

Runpfile Runp version

A unit can require a constraint on the Runp version.

This unit requires runp version greater the 0.5.0:

```
units:
  test1:
    description: test unit
    preconditions:
      runp:
        operator: GreaterThan
        version: 0.5.0
    host:
      command: env
      workdir: ${HOME}
```

The available operators are:

- `LessThan`
- `LessThanOrEqual`
- `Equal`
- `GreaterThanOrEqual`
- `GreaterThan`

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